SOFTWARE ENGINEERING LAB-01

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SECTION : B

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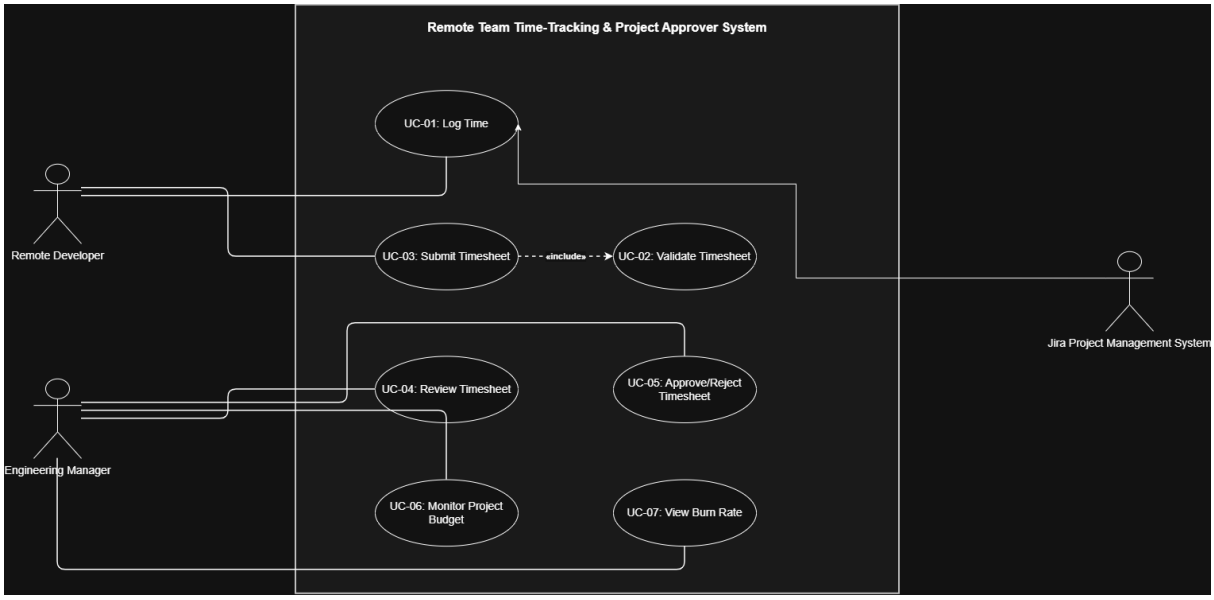
1) FUNCTIONAL REQUIREMENTS TABLE

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow remote developers to log time against specific project tasks and shall prevent timesheet entries from exceeding 24 hours in a single calendar day.	High	Pass: A valid timesheet is accepted and a timesheet exceeding 24 hours in one day is rejected. Fail: The system accepts more than 24 hours for one calendar day.	Ensures accurate and realistic tracking of developer working hours.
FR-002	Functional	The system shall allow developers to review and submit their completed timesheets to the Engineering Manager for approval.	High	Pass: A valid timesheet is successfully submitted and its status changes to Pending Approval. Fail: An invalid timesheet is submitted or the status does not change.	Provides a controlled workflow for timesheet submission and approval.
FR-003	Functional	The system shall allow the Engineering Manager to review submitted timesheets and either approve or reject them.	High	Pass: The manager can approve or reject a submitted timesheet and the system records the decision. Fail: The manager cannot complete or record the decision.	Enables managers to verify developer work before approval.
FR-004	Functional	The system shall provide project budget burn-rate information based on the hours recorded against project tasks.	Medium	Pass: The system displays updated project budget usage and burn-rate information. Fail: The displayed budget information is missing or inconsistent with recorded hours.	Helps managers monitor project spending and identify potential budget overruns.
FR-005	Functional	The system shall allow developers and Engineering Managers to view the current status and relevant details of timesheets according to their roles.	Medium	Pass: Users can view timesheet information permitted by their role. Fail: Required information is unavailable or unauthorized information is displayed.	Provides transparency while maintaining role-based access to timesheet information.

2) NON FUNCTIONAL REQUIREMENTS TABLE

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Timesheet approval status transitions and project budget burn charts shall update across dashboards within 500 ms under normal operating conditions.	High	Pass: Benchmarking confirms dashboard updates within 500 ms under the defined test load. Fail: Updates consistently exceed 500 ms.	Ensures managers receive timely information when monitoring approvals and project budgets.
NFR-002	Security	The system shall restrict timesheet and project information based on the user's assigned role, allowing developers to access their own records and managers to access records required for approval and project monitoring.	High	Pass: Role-based access tests allow only authorized operations and data access. Fail: A user can access or modify information outside their assigned permissions.	Protects employee timesheet information and prevents unauthorized modification of project data.

3) UML USE-CASE DIAGRAM



4) USE-CASE FLOW SPECIFICATION

Use Case: UC-03 - Submit Timesheet

Primary Actor: Remote Developer
Supporting Actor: Engineering Manager
Related Use Case: UC-02 - Validate Timesheet
Priority: High

Preconditions

1. The Remote Developer is logged into the Remote Team Time-Tracking & Project Approver System.
2. The developer has recorded working hours against one or more project tasks.
3. The timesheet contains the required project and time-entry information.

Postconditions

1. The timesheet is successfully submitted for managerial approval.
2. The timesheet status is changed to Pending Approval.
3. The Engineering Manager can review the submitted timesheet.

Main Success Scenario

1. The Remote Developer opens the timesheet containing the recorded work hours.
2. The system displays the recorded time entries and associated project tasks.
3. The Remote Developer reviews the timesheet and selects Submit Timesheet.
4. The system invokes UC-02 - Validate Timesheet to verify the submitted information.
5. The system confirms that all time entries are valid and that the total logged hours do not exceed 24 hours in a single calendar day.
6. The system submits the timesheet successfully.

7. The system changes the timesheet status to Pending Approval.
8. The system makes the submitted timesheet available to the Engineering Manager for review.
9. The use case ends successfully.

Alternate Flow

4a. Timesheet Validation Fails

1. During validation, the system detects an invalid timesheet entry.
2. The system identifies the reason for the validation failure and displays an appropriate error message.
3. The system does not submit the timesheet for approval.
4. The Remote Developer corrects the invalid entry.
5. The developer selects Submit Timesheet again.
6. The system validates the corrected timesheet.
7. If validation is successful, the system submits the timesheet and changes its status to Pending Approval.